

Math 550, S20  
Quiz 6

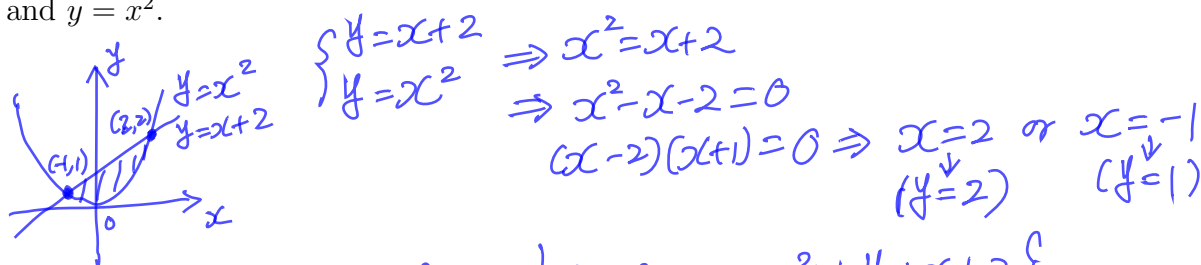
Name: Answer

**Instructions:** There are totally 4 problems. You must show all of your work to receive a credit for a correct answer.

**Problem 1.** (4 points) Calculate the iterated integral  $\int_0^1 \int_1^2 (4x^3 - 9x^2y^2) dy dx$

$$\begin{aligned} & \int_0^1 \int_1^2 4x^3 - 9x^2y^2 dy dx \\ &= \int_0^1 [4x^3y - 3x^2y^3]_{y=1}^{y=2} dx \\ &= \int_0^1 [(8x^3 - 24x^2) - (4x^3 - 3x^2)] dx \\ &= \int_0^1 4x^3 - 21x^2 dx \\ &= [x^4 - 7x^3]_{x=0}^{x=1} = (1 - 7) - (0) = -6 \end{aligned}$$

**Problem 2.** (6 points) Evaluate  $\iint_D x + 2y dA$ , where  $D$  is the region enclosed by  $y = x + 2$  and  $y = x^2$ .

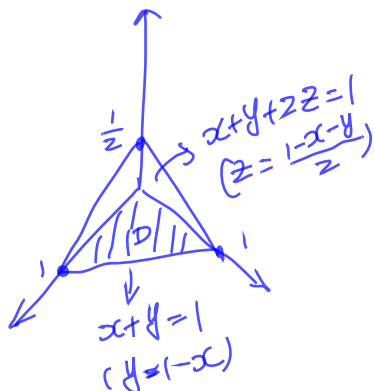


$$\begin{aligned} \begin{cases} y = x + 2 \\ y = x^2 \end{cases} &\Rightarrow x^2 = x + 2 \\ &\Rightarrow x^2 - x - 2 = 0 \\ &(x-2)(x+1) = 0 \Rightarrow x = 2 \text{ or } x = -1 \\ &\quad (y=2) \quad (y=1) \end{aligned}$$

$$\Rightarrow D = \{(x, y) \mid -1 \leq x \leq 2, x^2 \leq y \leq x + 2\}$$

$$\begin{aligned} \text{Thus} \\ \iint_D x + 2y dA &= \int_{-1}^2 \int_{x^2}^{x+2} x + 2y dy dx \\ &= \int_{-1}^2 [xy + y^2]_{y=x^2}^{y=x+2} dx \\ &= \int_{-1}^2 [x(x+2) + (x+2)^2] - [x^3 + x^4] dx \\ &= \int_{-1}^2 -x^4 - x^3 + 2x^2 + 6x + 4 dx \\ &= \left[ -\frac{x^5}{5} - \frac{x^4}{4} + \frac{2x^3}{3} + 3x^2 + 4x \right]_{x=-1}^{x=2} \\ &= \frac{333}{20} \end{aligned}$$

**Problem 3.** (5 points) Find the volume of the tetrahedra bounded by the planes  $y = 0$ ,  $z = 0$ ,  $x = 0$  and  $x + y + 2z = 1$ .



$$D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1-x\}$$

$$\text{Volume} = \iint_D z(x, y) dA$$

$$= \int_0^1 \int_0^{1-x} \frac{1-x-y}{2} dy dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{1-x} (1-x-y) dy dx$$

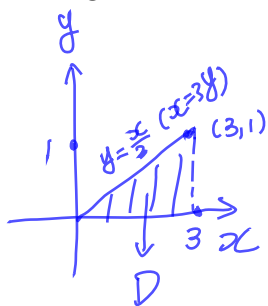
$$= \frac{1}{2} \int_0^1 \left[ (1-x)y - \frac{y^2}{2} \right]_{y=0}^{y=1-x} dx$$

$$= \frac{1}{2} \int_0^1 \frac{(1-x)^2}{2} dx$$

$$= -\frac{1}{12} (1-x)^3 \Big|_{x=0}^{x=1}$$

$$= \frac{1}{12}$$

**Problem 4.** (5 points) Given a double integral  $\int_0^1 \int_{3y}^3 \sin(x^2) dx dy$ . Reverse the order of the integration and evaluate the resulted iterated integral.



$$D = \{(x, y) \mid 0 \leq y \leq 1, 3y \leq x \leq 3\} \rightarrow \text{TYPE II}$$

$$= \{(x, y) \mid 0 \leq x \leq 3, 0 \leq y \leq \frac{x}{3}\} \rightarrow \text{TYPE I}$$

$$\int_0^1 \int_{3y}^3 \sin x^2 dx dy = \iint_D \sin x^2 dA$$

$$= \int_0^3 \int_0^{\frac{x}{3}} \sin x^2 dy dx$$

$$= \int_0^3 (\sin x^2) y \Big|_{y=0}^{y=\frac{x}{3}} dx$$

$$= \int_0^3 \frac{x}{3} \sin x^2 dx$$

$$= \left[ -\frac{1}{6} \cos x^2 \right]_{x=0}^{x=3} = \frac{1}{6} (1 - \cos 9)$$